Software Requirements Specification (SRS)

Project: Helping Hands (HH), an app that lets management staff of a retail chain volunteer to work in stores during very busy times such as weekends and holidays

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	28-10-2026	Aamer Khan, Bhuvan Kumar P	Complete SRS with use-case diagrams and RTM	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Guide			

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification for Helping Hands, a web application for a retail chain. When stores expect heavy footfall, such as on weekends, public holidays and sale days, management staff offer to work a shift in a store. Helping Hands lets stores ask for that help and lets managers volunteer for it. The document defines the functional and non-functional requirements, interfaces and acceptance criteria for the developers, testers and evaluators of the project.

1.2 Scope

Helping Hands covers: setting up stores, busy periods and volunteering rules; posting help requests (shifts) for a store; browsing and signing up for shifts; approving or declining volunteers; withdrawing before a cutoff; in-app notifications and reminders; recording attendance and hours; coverage views and reports; and an audit trail. It is a web prototype for one retail chain.

Out of scope: pay, overtime and expense claims for volunteered hours, integration with the chain's HR, payroll or rostering systems, and SMS or push notifications. Only in-app notifications are built in this version.

1.3 Audience

Developers, QA engineers, store and management staff who act as users, the project guide and the course evaluators.

1.4 Definitions

Term	Meaning
Busy period	A date range when stores expect heavy footfall, for example a weekend, a public holiday or a sale event.
Help request (shift)	A store's request for a number of volunteers for one task on one date, between a start time and an end time.
Management volunteer	A member of management staff who offers to work a shift in a store.
Slot	One volunteer place on a shift. A shift needing three volunteers has three slots.
Sign-up	A volunteer's request to work a shift. Its status is Requested, Confirmed, Declined or Withdrawn.
Coverage	For a shift or busy period, the slots needed compared with the slots requested and confirmed.
Withdrawal cutoff	The latest time before a shift starts when a volunteer may still withdraw (default 24 hours).
Minimum notice	The shortest time before a shift starts when a volunteer may still sign up (default 12 hours).
Weekly hour limit	The most volunteering hours one person may hold in a calendar week (default 8 hours).
RBAC	Role-based access control: what a user can do depends on their role.

2. Overall description

2.1 Product perspective

Helping Hands is a new, self-contained web application. Staff use it from a laptop or phone browser. A React web app talks to a REST API, which runs the business rules and stores data in a relational database. The API checks every sign-up against the shift's free slots and the volunteer's other commitments before saving it, and tells the right people through in-app notifications.

2.2 Major product functions

- Sign users in and apply role-based access, with store managers limited to their own store.
- Let administrators manage users, stores, busy periods and volunteering rules.
- Let store managers post, edit and cancel help requests for their store.
- Let volunteers browse and filter open shifts and sign up for them.
- Check each sign-up for free slots, minimum notice, clashing shifts and the weekly hour limit.
- Let store managers approve or decline sign-ups without ever overbooking a shift.
- Let volunteers withdraw before the cutoff, which frees the slot again.
- Send in-app notifications and a reminder 24 hours before each confirmed shift.
- Record attendance and hours, show coverage per busy period, and report hours contributed.

2.3 User roles and characteristics

Role	Characteristics and permissions
Administrator (operations or HR)	Manages users, roles and stores; defines busy periods and volunteering rules; reads all reports and the audit history. Uses a laptop.
Store manager	Posts and cancels help requests for their own store; approves or declines sign-ups; records attendance; views coverage. Often busy on the shop floor, so screens must be quick.
Management volunteer	Head-office or regional manager who browses shifts, signs up and withdraws, and checks their schedule and hours. Mostly uses a phone.

2.4 Operating environment

Current versions of Chrome, Edge, Safari and Firefox on laptops and phones (screens 360 px wide and up). The server runs Python 3.12 with FastAPI and PostgreSQL 16 (SQLite in local development), and the test environment is built with Docker.

2.5 Constraints

- Every sign-up decision must respect the slot count of the shift, even when two managers act at the same moment.
- Shift times are entered and shown in the store's local time.
- Volunteer hours are recorded for recognition and reporting only; they are not linked to pay.
- The project is built by a four-member team within two sprints, so integrations are left for later versions.

2.6 Assumptions and dependencies

- Every volunteer already has a staff email address and an account created by the administrator.
- Each store has one store manager account in this version.
- The default rules (8-hour weekly limit, 12-hour minimum notice, 24-hour withdrawal cutoff) are acceptable until the chain decides otherwise.

3. External interface requirements

3.1 User interfaces

- Volunteer home: open shifts grouped by date, with store, task, time and free slots, plus filters and a one-tap Sign up button.
- My Schedule: upcoming and past shifts with status and hours, and a Withdraw button until the cutoff.
- Store manager dashboard: shifts per busy period with slots needed, requested and confirmed, and pending sign-ups to approve.
- Admin pages for users, stores, busy periods, rules, reports and audit history.
- Statuses such as Full, Cancelled or Confirmed are written in text, not shown by colour alone.

3.2 Hardware interfaces

None. The system needs no special hardware; users bring their own laptop or phone.

3.3 Software interfaces

- REST API under /api/v1, JSON over HTTPS, used by the web app.
- PostgreSQL for all application data, reached only through the data-access layer.
- A scheduled job inside the server sends the 24-hour reminders.

3.4 Communications

- HTTPS with TLS 1.2 or higher for all traffic when deployed; localhost during development.
- Signed session tokens in the Authorization header on every API call.
- Notifications are stored in the database and shown in the app; no outside messaging service is used.

4. System features (detailed)

Each requirement below has acceptance criteria and a reference test case. IDs follow HH-F-###. In total the project defines 23 functional requirements, 7 non-functional requirements, 3 security objectives and 8 security requirements.

4.1 User access and store setup

Description: Sign users in, apply role-based access, and let the administrator set up stores, busy periods and volunteering rules.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-001	The system shall let a registered user sign in with email and password and sign out.	Functional	High	All users	AC-HH-F-001: Valid credentials start a session; a wrong password is rejected with a general message. Test: TC-AUTH-01, TC-AUTH-02	Auth and RBAC
HH-F-002	The system shall restrict every action by role, and limit store managers to their own store's data.	Functional	High	Security	AC-HH-F-002: A volunteer cannot post shifts; a manager cannot change another store's shift (403). Test: TC-AUTH-03, TC-AUTH-04	Depends on HH-F-001
HH-F-003	The system shall let an administrator create users, change their role and deactivate them.	Functional	High	Administrator	AC-HH-F-003: A new store manager account is active with the right role; a deactivated user cannot sign in. Test: TC-USER-01	Auth and RBAC
HH-F-004	The system shall let an administrator create, edit and deactivate stores and assign one store manager to each.	Functional	High	Administrator	AC-HH-F-004: The store is listed with its manager; a deactivated store accepts no new shifts. Test: TC-STORE-01	Store and Busy Period
HH-F-005	The system shall let an administrator define busy periods with a name, start date and end date.	Functional	High	Operations	AC-HH-F-005: A saved busy period can be chosen when posting a shift. Test: TC-BP-01	Store and Busy Period
HH-F-006	The system shall let an administrator set the weekly hour limit, minimum notice and withdrawal cutoff.	Functional	Medium	Administrator / HR policy	AC-HH-F-006: New values apply to sign-ups made after the change. Test: TC-RULE-01	Used by HH-F-012, 013, 016

4.2 Help requests (shifts)

Description: Let store managers ask for help for a busy period, and keep every shift valid.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-007	The system shall let a store manager post a help request with date, start and end time, task, number of volunteers needed, busy period and notes.	Functional	High	Store manager	AC-HH-F-007: The shift is saved as Open and shown to volunteers with its free slots. Test: TC-SHIFT-01	Shift Service

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-008	The system shall reject a shift whose end time is not after its start time, whose date has passed, or which needs fewer than one volunteer.	Functional	High	Store manager	AC-HH-F-008: Each invalid case gets a clear message and nothing is saved. Test: TC-SHIFT-02, TC-SHIFT-03	Server-side validation
HH-F-009	The system shall let a store manager edit or cancel their store's shifts, and notify every volunteer signed up to a cancelled shift.	Functional	High	Store manager / volunteers	AC-HH-F-009: The shift becomes Cancelled and each affected volunteer gets a notification. Test: TC-SHIFT-04	Notification Service

4.3 Finding and signing up for shifts

Description: Help volunteers find a suitable shift quickly and stop sign-ups that would break the rules.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-010	The system shall let volunteers browse open shifts and filter them by date, busy period, store and task.	Functional	High	Management volunteer	AC-HH-F-010: Only matching shifts appear, each with its correct number of free slots. Test: TC-BROWSE-01	Shift Service
HH-F-011	The system shall let a volunteer sign up for an open shift, creating a Requested sign-up and notifying the store manager.	Functional	High	Management volunteer	AC-HH-F-011: The sign-up is Requested and the manager sees it in the pending list. Test: TC-SIGN-01	Sign-up Service
HH-F-012	The system shall reject a sign-up when the shift is full, cancelled or already started, when it is inside the minimum notice, or when it overlaps another shift the volunteer holds.	Functional	High	Store manager / volunteer	AC-HH-F-012: Each case is refused with its reason; shifts that only touch end to start are allowed. Test: TC-SIGN-02 to TC-SIGN-05	Rules from HH-F-006
HH-F-013	The system shall reject a sign-up that would take the volunteer over the weekly hour limit.	Functional	Medium	HR policy	AC-HH-F-013: The sign-up is refused and the hours still available are shown. Test: TC-SIGN-06	Rules from HH-F-006

4.4 Approval and withdrawal

Description: Let store managers choose their volunteers, and let volunteers step back in good time.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-014	The system shall let a store manager approve or decline sign-ups for their store's shifts, and never confirm more volunteers than the shift needs.	Functional	High	Store manager	AC-HH-F-014: Approval confirms the volunteer; an approval beyond the slot count is refused with 409. Test: TC-APPR-01 to TC-APPR-03, TC-CONC-01	One database transaction (HH-NF-002)
HH-F-015	The system shall mark a shift Filled when all its slots are confirmed and Open again when a slot is freed.	Functional	Medium	Store manager / volunteer	AC-HH-F-015: Shift status always follows the confirmed count. Test: TC-APPR-01, TC-STAT-01	Shift Service
HH-F-016	The system shall let a volunteer withdraw a sign-up before the withdrawal cutoff, free the slot and notify the store manager.	Functional	Medium	Management volunteer	AC-HH-F-016: Before the cutoff the sign-up becomes Withdrawn; after it, withdrawal is refused. Test: TC-WD-01, TC-WD-02	Cutoff from HH-F-006

4.5 Notifications and schedules

Description: Keep everyone informed without relying on email.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-017	The system shall create an in-app notification for each new sign-up, approval, decline, withdrawal and cancellation.	Functional	Medium	All users	AC-HH-F-017: Each event creates one notification for the right user. Test: TC-NOTE-01	Notification Service
HH-F-018	The system shall remind each confirmed volunteer 24 hours before their shift starts.	Functional	Medium	Management volunteer	AC-HH-F-018: One reminder per confirmed sign-up, sent once. Test: TC-NOTE-02	Scheduled job
HH-F-019	The system shall show each volunteer their upcoming and past shifts with status and hours.	Functional	Medium	Management volunteer	AC-HH-F-019: My Schedule lists only that volunteer's sign-ups. Test: TC-SCHED-01	Sign-up Service

4.6 Attendance, coverage and reports

Description: Record who turned up and show how well each busy period was covered.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HH-F-020	The system shall let a store manager record each confirmed volunteer as attended or no-show, with hours worked between 0 and the shift length.	Functional	High	Store manager	AC-HH-F-020: Attendance and hours are saved and appear in the volunteer's history; invalid hours are refused. Test: TC-ATT-01, TC-ATT-02	Attendance Service
HH-F-021	The system shall show a store manager the coverage of their store's shifts for each busy period: slots needed, requested and confirmed.	Functional	Medium	Store manager	AC-HH-F-021: Coverage figures match the stored shifts and sign-ups. Test: TC-COV-01	Reporting Service
HH-F-022	The system shall produce reports of volunteer hours by volunteer, store and busy period, coverage across stores, and no-shows, with CSV export.	Functional	Medium	Administrator / store manager	AC-HH-F-022: Report totals match recorded attendance for the chosen period. Test: TC-REP-01	Reporting Service
HH-F-023	The system shall record an audit event for every change to shifts, sign-up decisions, attendance and user roles.	Functional	High	Administrator	AC-HH-F-023: Each entry shows the action, the time and the acting user. Test: TC-AUDIT-01	Audit Service

5. Non-functional requirements (detailed)

The NFRs below are measurable and tied to the test plan. IDs follow HH-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HH-NF-001	With 2,000 shifts and 500 users, the open-shift list and the coverage view shall load within 2 seconds for at least 90% of requests at 50 concurrent users.	Performance	High	JMeter run of 20 or more requests; 90th percentile under 2 s. Test: TC-PERF-01
HH-NF-002	Two approvals for the last free slot of a shift made at the same moment shall never both succeed.	Data integrity	High	Concurrent approval test confirms exactly one. Test: TC-CONC-01
HH-NF-003	Committed data shall survive an application restart, and the deployed database shall be backed up daily.	Reliability	High	Restart test; backup job log. Test: TC-REL-01
HH-NF-004	A volunteer shall be able to find and sign up for a shift in three steps or fewer from the home page on a 360 px phone screen.	Usability	High	Timed walkthrough by a tester. Test: TC-UX-01
HH-NF-005	Core pages shall meet WCAG 2.1 AA: keyboard-operable forms, readable contrast, and statuses in text.	Accessibility	Medium	Keyboard run and automated check. Test: TC-UX-02

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HH-NF-006	Business rules shall sit in a service layer separate from the API and UI, with at least 70% line coverage from unit tests.	Maintainability	Medium	CI coverage report. Test: TC-DEV-01
HH-NF-007	A new team member shall be able to set up and run the project from the README on a clean machine.	Portability	Medium	Clean-machine setup run. Test: TC-DEV-02

5.1 Security

5.1.1 Security objectives

- SO-1: Only the right people can change shifts, sign-ups and attendance, according to their role and store.
- SO-2: Staff personal data, such as contact details and schedules, is seen only by those who need it.
- SO-3: Every sign-up decision and every recorded hour can be traced to who made it and when.

5.1.2 Security requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
HH-SR-001	The system shall use TLS 1.2 or higher for all traffic when deployed, and redirect HTTP to HTTPS.	Security	High	HTTP is redirected; TLS 1.0 is refused. Test: TC-SEC-07
HH-SR-002	The system shall store passwords as salted bcrypt hashes and lock an account for 15 minutes after 5 failed sign-ins.	Security	High	No plaintext passwords stored; lockout works. Test: TC-SEC-01, TC-SEC-02
HH-SR-003	The system shall end a session after 30 minutes of inactivity and invalidate the token on sign-out.	Security	High	An idle user must sign in again. Test: TC-SEC-03
HH-SR-004	The system shall check role and store ownership on the server for every protected request, denying by default.	Security	High	Direct calls outside a user's role or store get 403. Test: TC-AUTH-03, TC-AUTH-04
HH-SR-005	The system shall validate all input on the server and use parameterized queries only.	Security	High	Malformed or hostile input is rejected safely. Test: TC-SEC-04
HH-SR-006	The system shall show a volunteer's phone and email only to the manager of a shift they are confirmed for, and to administrators.	Privacy	High	Other volunteers cannot see the contact details. Test: TC-SEC-05
HH-SR-007	The system shall keep an append-only audit log with no passwords or tokens in it.	Security	High	Audit entries cannot be edited and hold no secrets. Test: TC-AUDIT-01
HH-SR-008	The system shall rate-limit sign-in and sign-up requests per user to stop request floods when a busy period opens.	Security	Medium	Requests over the limit get 429. Test: TC-SEC-06

6. Quality attributes & acceptance tests

- Exit criteria for acceptance: all high-priority functional and security requirements are implemented and verified, no critical or high defect is open, HH-NF-001 to HH-NF-003 are met, and the RTM shows every test case passed.

- Acceptance test suites: access and setup, help requests, sign-up and eligibility, approval and withdrawal, notifications, attendance and reports, performance, security, usability and accessibility.
- Integrity check: after a scripted run of sign-ups, approvals and withdrawals, no shift has more confirmed volunteers than slots, and hours in the report equal the recorded attendance (TC-CONC-01, TC-REP-01).

7. System models and diagrams

7.1 UML use-case diagrams

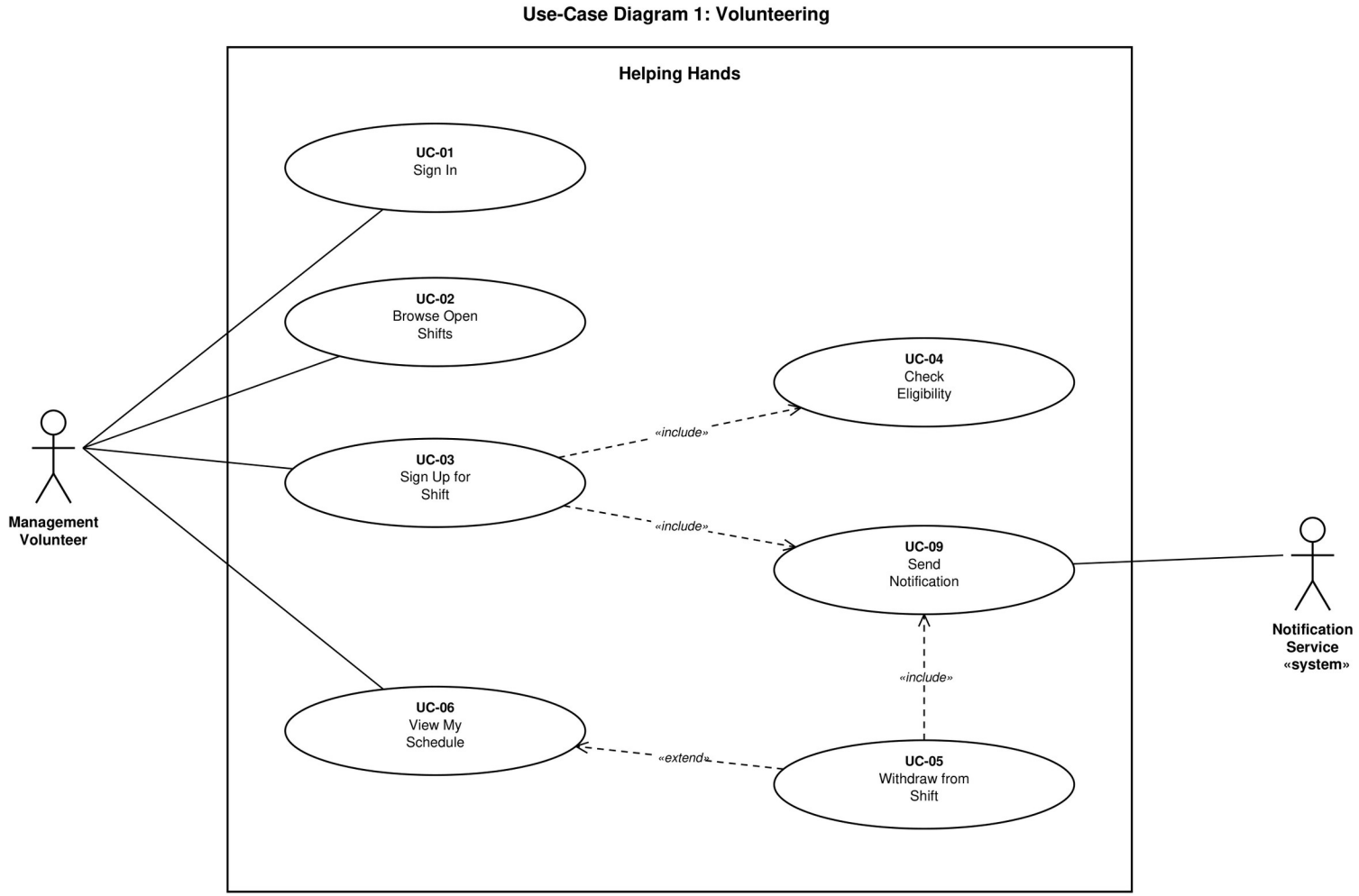


Figure 1. Use-case diagram 1: volunteering (management volunteer)

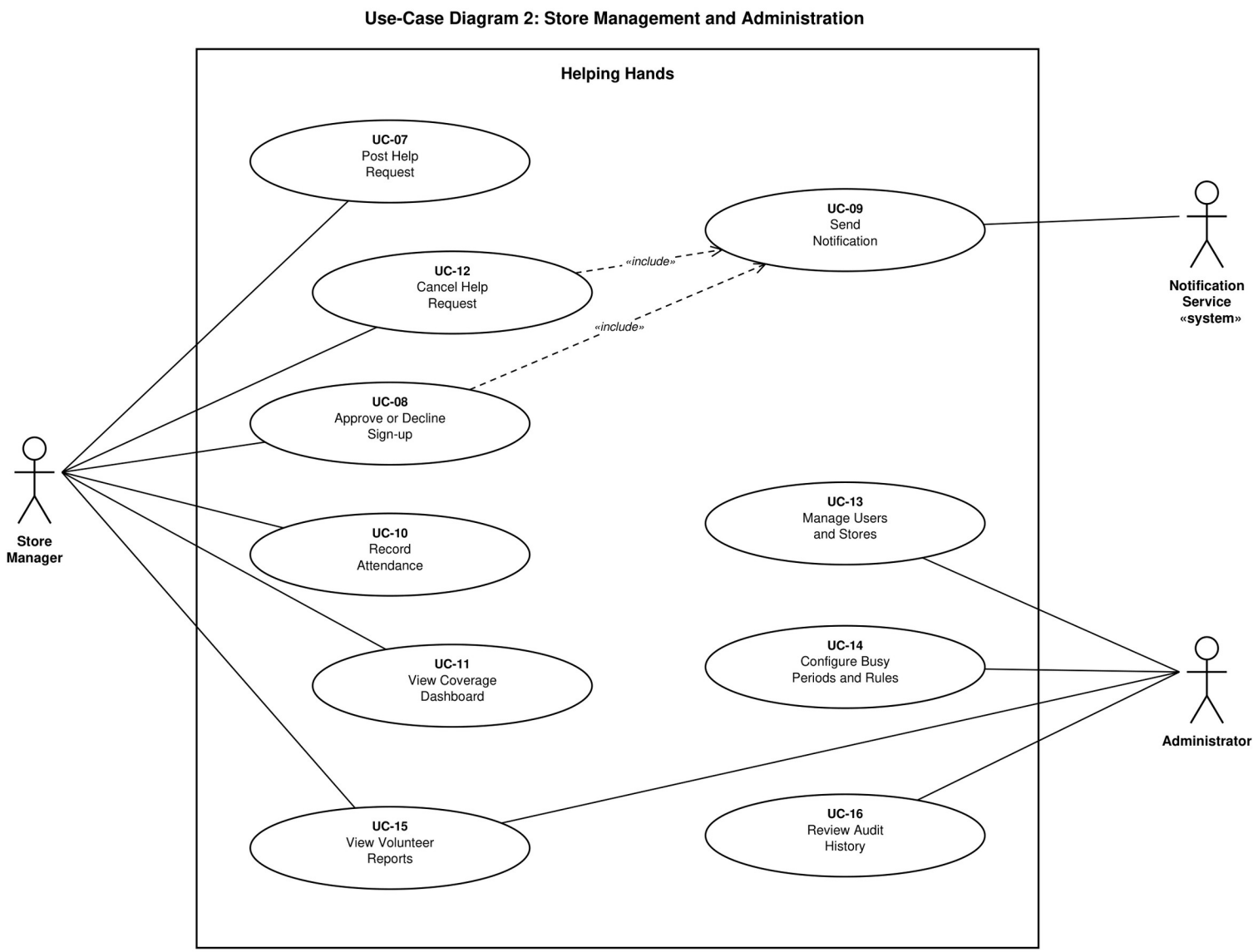


Figure 2. Use-case diagram 2: store management and administration

7.2 Use-case summary

Use case	Primary actor	Related requirements
UC-01 Sign In	All users	HH-F-001, HH-F-002, HH-SR-002
UC-02 Browse Open Shifts	Management volunteer	HH-F-010
UC-03 Sign Up for Shift	Management volunteer	HH-F-011
UC-04 Check Eligibility (included)	System	HH-F-012, HH-F-013
UC-05 Withdraw from Shift	Management volunteer	HH-F-016
UC-06 View My Schedule	Management volunteer	HH-F-019
UC-07 Post Help Request	Store manager	HH-F-007, HH-F-008
UC-08 Approve or Decline Sign-up	Store manager	HH-F-014, HH-F-015
UC-09 Send Notification	Notification Service (system)	HH-F-017, HH-F-018
UC-10 Record Attendance	Store manager	HH-F-020
UC-11 View Coverage Dashboard	Store manager	HH-F-021
UC-12 Cancel Help Request	Store manager	HH-F-009
UC-13 Manage Users and Stores	Administrator	HH-F-003, HH-F-004
UC-14 Configure Busy Periods and Rules	Administrator	HH-F-005, HH-F-006
UC-15 View Volunteer Reports	Store manager, Administrator	HH-F-022
UC-16 Review Audit History	Administrator	HH-F-023

8. Requirements Traceability Matrix (RTM)

Status: N = Not started, P = In progress, A = Accepted (tested and passed).

Req ID	Requirement short	Section ref / Design spec	Module	Test case(s)	Status (N/P/A)	Comments
HH-F-001	Sign in and out	4.1 / SAD 4.3	Auth and RBAC	TC-AUTH-01, 02	N	UC-01
HH-F-002	Role and store access	4.1 / SAD 3.9	Auth and RBAC	TC-AUTH-03, 04	N	UC-01
HH-F-003	Manage users	4.1 / SAD 3.4	Auth and RBAC	TC-USER-01	N	UC-13
HH-F-004	Manage stores	4.1 / SAD 3.4	Store and Busy Period	TC-STORE-01	N	UC-13
HH-F-005	Busy periods	4.1 / SAD 3.4	Store and Busy Period	TC-BP-01	N	UC-14
HH-F-006	Volunteering rules	4.1 / SAD 3.4	Store and Busy Period	TC-RULE-01	N	UC-14
HH-F-007	Post help request	4.2 / SAD 4.3	Shift Service	TC-SHIFT-01	N	UC-07
HH-F-008	Validate shift	4.2 / SAD 4.3	Shift Service	TC-SHIFT-02, 03	N	UC-07
HH-F-009	Cancel and notify	4.2 / SAD 3.4	Shift, Notification	TC-SHIFT-04	N	UC-12
HH-F-010	Browse and filter	4.3 / SAD 4.3	Shift Service	TC-BROWSE-01	N	UC-02
HH-F-011	Sign up	4.3 / SAD 4.2	Sign-up Service	TC-SIGN-01	N	UC-03
HH-F-012	Reject ineligible sign-up	4.3 / SAD 4.2	Sign-up Service	TC-SIGN-02 to 05	N	UC-04

Req ID	Requirement short	Section ref / Design spec	Module	Test case(s)	Status (N/P/A)	Comments
HH-F-013	Weekly hour limit	4.3 / SAD 4.2	Sign-up Service	TC-SIGN-06	N	UC-04
HH-F-014	Approve without overbooking	4.4 / SAD 4.2	Sign-up Service	TC-APPR-01 to 03, TC-CONC-01	N	UC-08
HH-F-015	Filled / Open status	4.4 / SAD 3.4	Shift Service	TC-STAT-01	N	UC-08
HH-F-016	Withdraw before cutoff	4.4 / SAD 3.4	Sign-up Service	TC-WD-01, 02	N	UC-05
HH-F-017	In-app notifications	4.5 / SAD 3.4	Notification Service	TC-NOTE-01	N	UC-09
HH-F-018	24-hour reminder	4.5 / SAD 3.6	Notification Service	TC-NOTE-02	N	UC-09
HH-F-019	My schedule	4.5 / SAD 3.4	Sign-up Service	TC-SCHED-01	N	UC-06
HH-F-020	Record attendance	4.6 / SAD 3.4	Attendance Service	TC-ATT-01, 02	N	UC-10
HH-F-021	Coverage view	4.6 / SAD 3.4	Reporting Service	TC-COV-01	N	UC-11
HH-F-022	Hours reports	4.6 / SAD 3.4	Reporting Service	TC-REP-01	N	UC-15
HH-F-023	Audit events	4.6 / SAD 3.9	Audit Service	TC-AUDIT-01	N	UC-16
HH-NF-001	Shift list under 2 s	5 / SAD 3.6	REST API, database	TC-PERF-01	N	
HH-NF-002	No overbooking	5 / SAD 3.5	Sign-up Service	TC-CONC-01	N	
HH-NF-003	Restart and backup	5 / SAD 3.7	Database	TC-REL-01	N	
HH-NF-004	Sign up in 3 steps	5 / SAD 4.5	Web App	TC-UX-01	N	
HH-NF-005	WCAG 2.1 AA	5 / SAD 4.5	Web App	TC-UX-02	N	
HH-NF-006	Layered, 70% coverage	5 / SAD 3.5	All services	TC-DEV-01	N	
HH-NF-007	README setup	5 / SAD 3.6	Repository	TC-DEV-02	N	
HH-SR-001	TLS only	5.1.2 / SAD 3.9	Deployment	TC-SEC-07	N	
HH-SR-002	Hashing and lockout	5.1.2 / SAD 3.9	Auth and RBAC	TC-SEC-01, 02	N	
HH-SR-003	Session expiry	5.1.2 / SAD 3.9	Auth and RBAC	TC-SEC-03	N	
HH-SR-004	Server-side access checks	5.1.2 / SAD 3.9	Auth and RBAC	TC-AUTH-03, 04	N	
HH-SR-005	Input validation	5.1.2 / SAD 3.9	REST API	TC-SEC-04	N	
HH-SR-006	Contact privacy	5.1.2 / SAD 3.9	Sign-up Service	TC-SEC-05	N	
HH-SR-007	Append-only audit	5.1.2 / SAD 3.9	Audit Service	TC-AUDIT-01	N	
HH-SR-008	Rate limiting	5.1.2 / SAD 3.9	REST API	TC-SEC-06	N	